📄 Report: CI/CD Pipeline Using Jenkins and Docker

1 Aim

To implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline using **Jenkins** and **Docker** for building and pushing a Python-based application image to DockerHub.

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2 Tools Used

- **Jenkins** – Automation server for CI/CD
- **Docker** – Containerization platform
- **DockerHub** – Image registry
- **GitHub** – Source code repository
- **Python** – Application programming language

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3 System Requirements

- Windows/Linux system
- Docker Desktop installed and running
- Jenkins installed (running on port 8080)
- GitHub repository containing project files
- DockerHub account

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4 Project Structure

The project contains:

- `main.py`
- `Student.py`
- `Faculty.py`
- `Course.py`
- `Fees.py`
- `Alumni.py`
- `Dockerfile`

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5 Dockerfile Configuration

```
FROM python:3.10-slim

WORKDIR /app

COPY . .

RUN if [ -f requirements.txt ]; then pip install --no-cache-dir -r
requirements.txt; fi

CMD ["python", "main.py"]
```

Explanation:

- Base image: Python 3.10
- Working directory set to `/app`
- Copies project files
- Installs dependencies (if available)
- Runs `main.py`

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6 Jenkins Configuration Steps

Step 1: Install Required Plugins

Navigate to:

```
Manage Jenkins → Plugins
```

Install:

- Docker Pipeline Plugin
- Git Plugin

Restart Jenkins.

Step 2: Add DockerHub Credentials

Navigate to:

```
Manage Jenkins → Credentials → Global → Add Credentials
```

Select:

- Kind: Username with Password
- Username: DockerHub Username

- Password: DockerHub Password
- ID: dockerhub-cred

Save credentials.

Step 3: Create Pipeline Job

1. Click **New Item**
2. Enter job name
3. Select **Pipeline**
4. Click OK

Step 4: Add Jenkins Pipeline Script

```
pipeline {
    agent any

    environment {
        DOCKER_USERNAME = "your_dockerhub_username"
        IMAGE_NAME = "college-system"
        IMAGE_TAG = "latest"
    }

    stages {

        stage('Checkout Code') {
            steps {
                git branch: 'main',
                    url: 'https://github.com/DivyanshSingh8899/College-
Management-System-Python.git'
            }
        }

        stage('Verify Docker') {
            steps {
                bat 'docker --version'
            }
        }

        stage('Build Docker Image') {
            steps {
                bat 'docker build -t
%DOCKER_USERNAME%/%IMAGE_NAME%:%IMAGE_TAG% .'
            }
        }

        stage('Login to DockerHub') {
            steps {
                script {
                    withCredentials([usernamePassword(
                        credentialsId: 'dockerhub-cred',
                        usernameVariable: 'DOCKER_USER',
                        passwordVariable: 'DOCKER_PASS'
```

```
                    )]) {
                        bat '''
                        echo %DOCKER_PASS% | docker login -u %DOCKER_USER%
--password-stdin
                        '''
                    }
                }
            }
        }

        stage('Push Image') {
            steps {
                bat 'docker push
%DOCKER_USERNAME%/%IMAGE_NAME%:%IMAGE_TAG%'
            }
        }
    }
}
```

---

7 Execution Steps

1. Click **Build Now**
2. Jenkins performs:
    - Code checkout from GitHub
    - Docker verification
    - Image build
    - DockerHub login
    - Image push to DockerHub

---

8 Output

- Docker image successfully built
- Image pushed to DockerHub repository
- CI/CD pipeline executed successfully

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9 Result

The CI/CD pipeline using Jenkins and Docker was successfully implemented. The Python application was containerized and pushed to DockerHub automatically through Jenkins automation.

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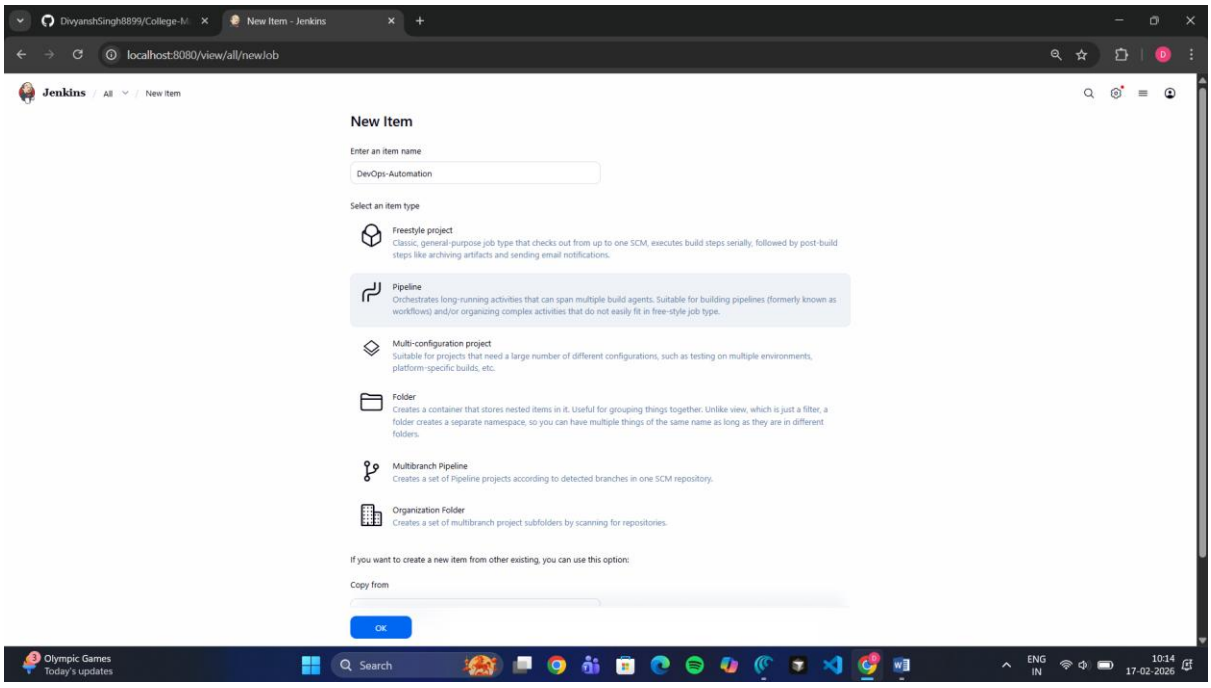
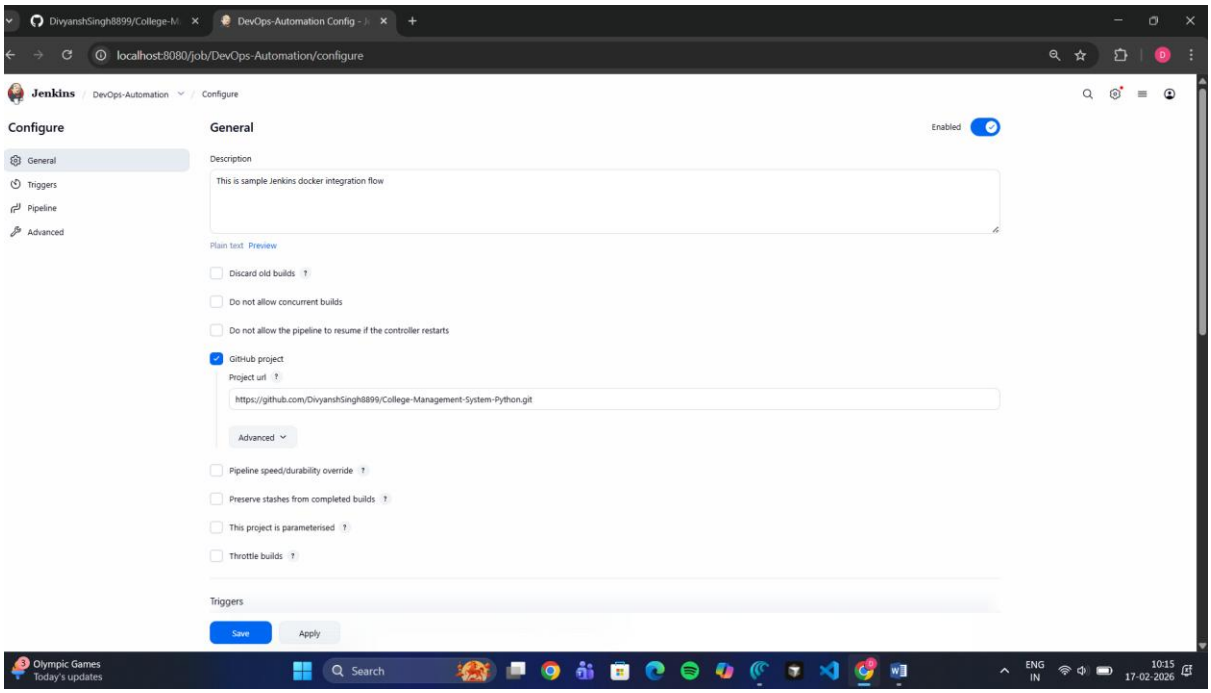
10 Conclusion

This experiment demonstrates how Jenkins can automate the build and deployment process using Docker, ensuring efficient continuous integration and delivery.

Screenshots:-

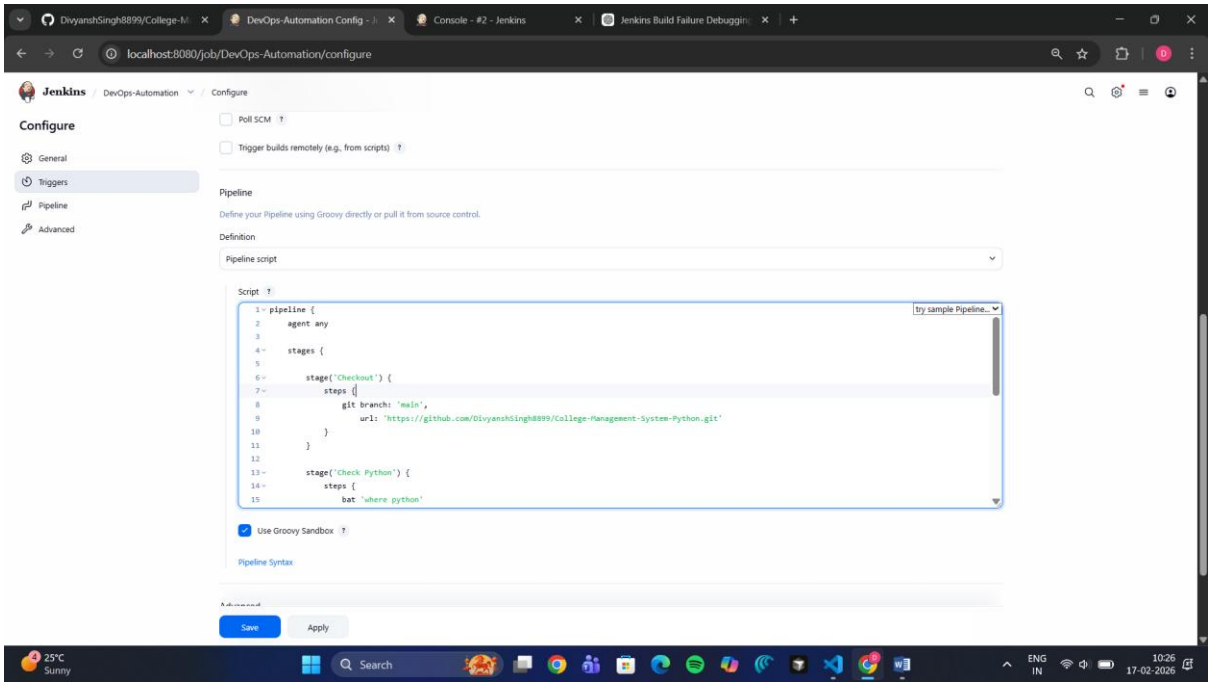


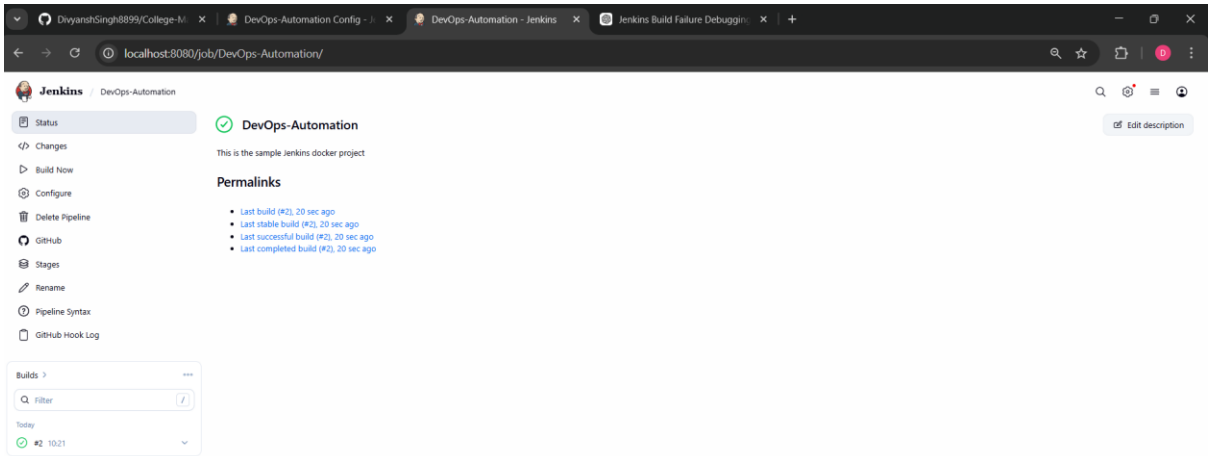


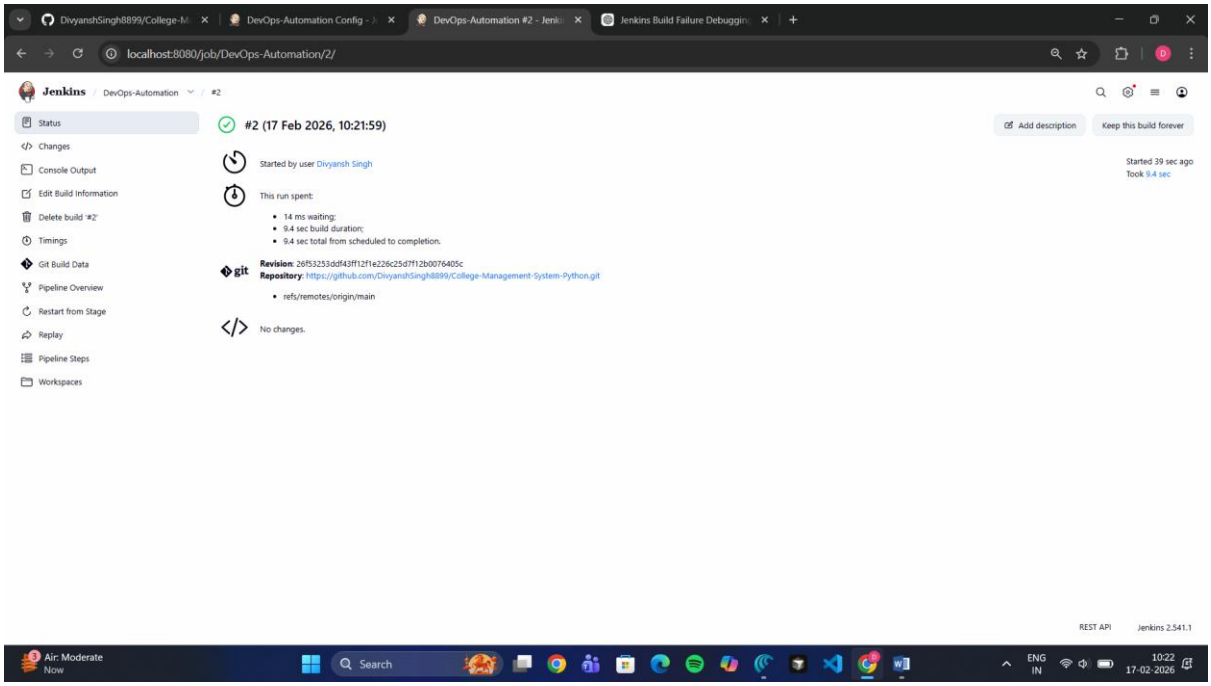
Triggers

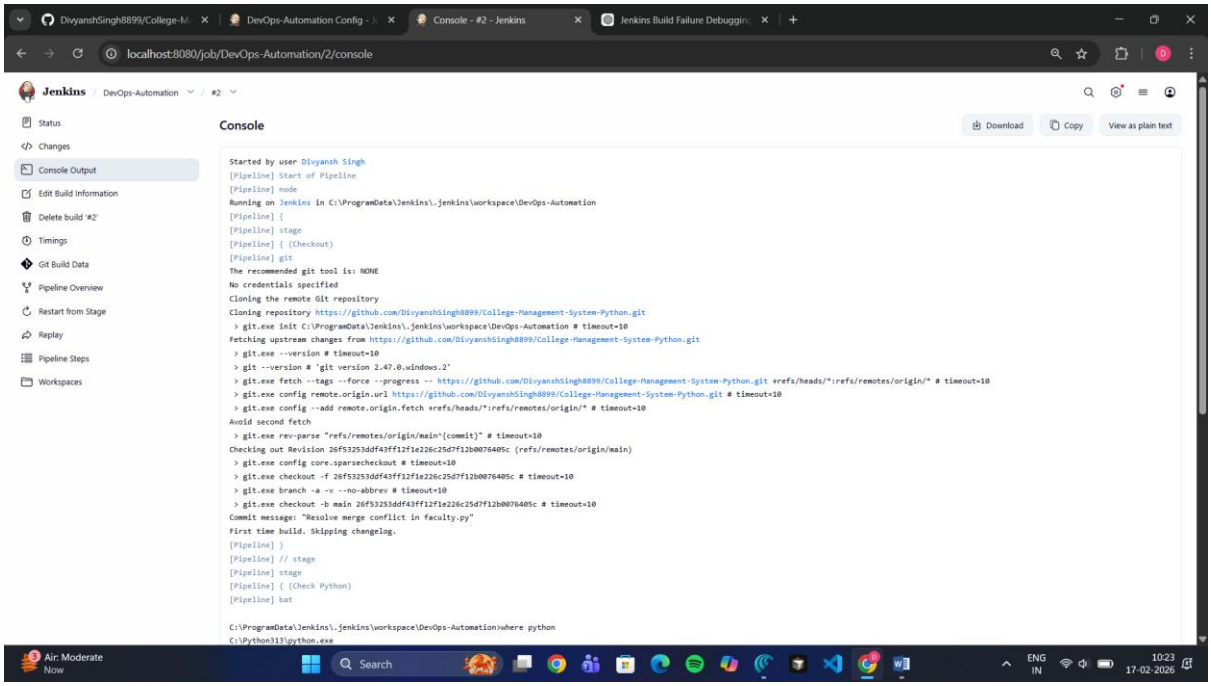
Set up automated actions that start your build based on specific events, like code changes or scheduled times.

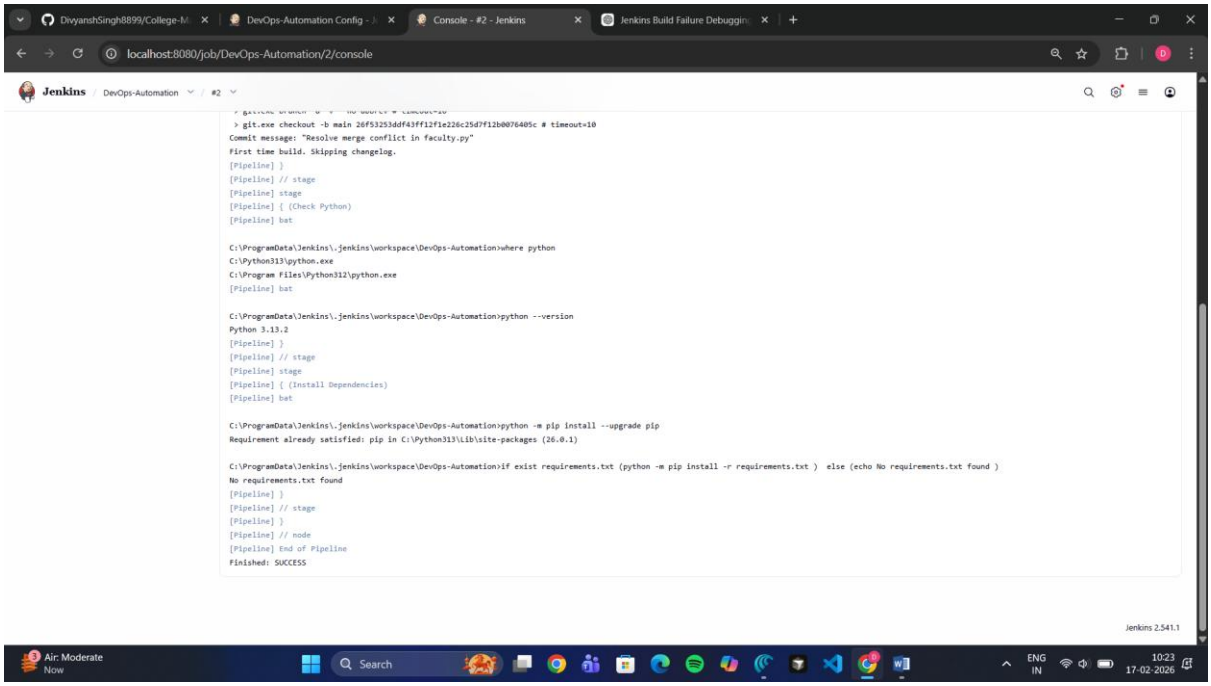
- [ ] Build after other projects are built ?
- [ ] Build periodically ?
- [x] GitHub hook trigger for GITScm polling ?
- [ ] Poll SCM ?
- [ ] Trigger builds remotely (e.g., from scripts) ?

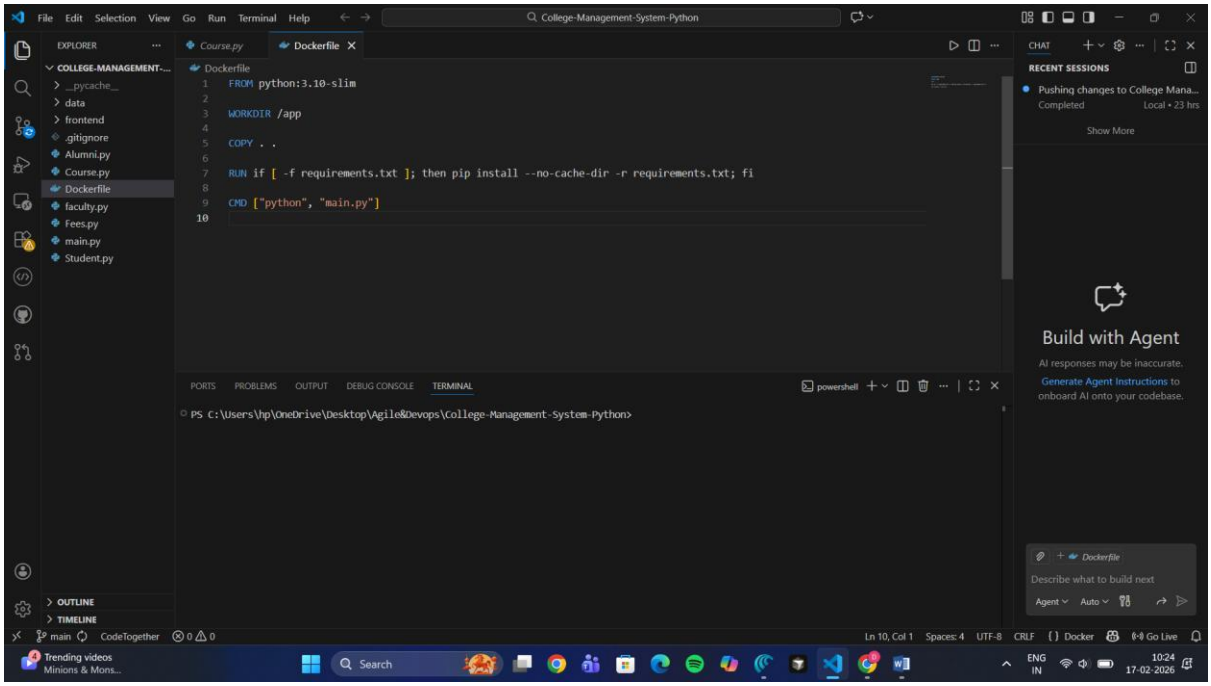


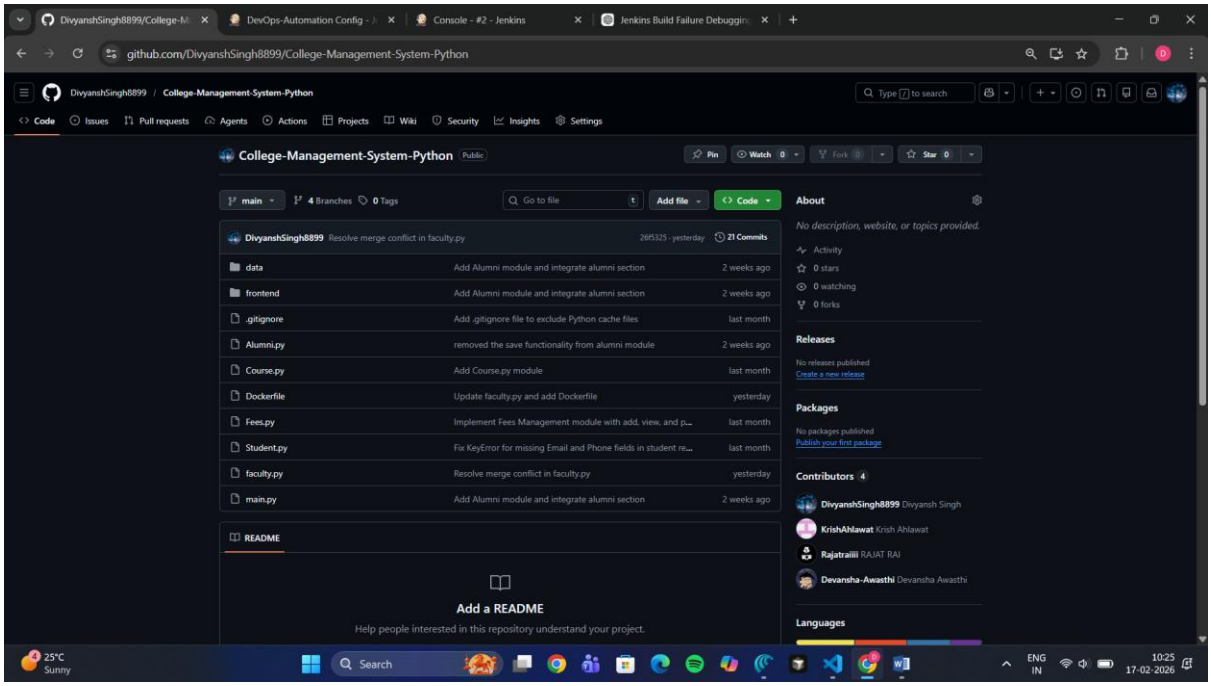


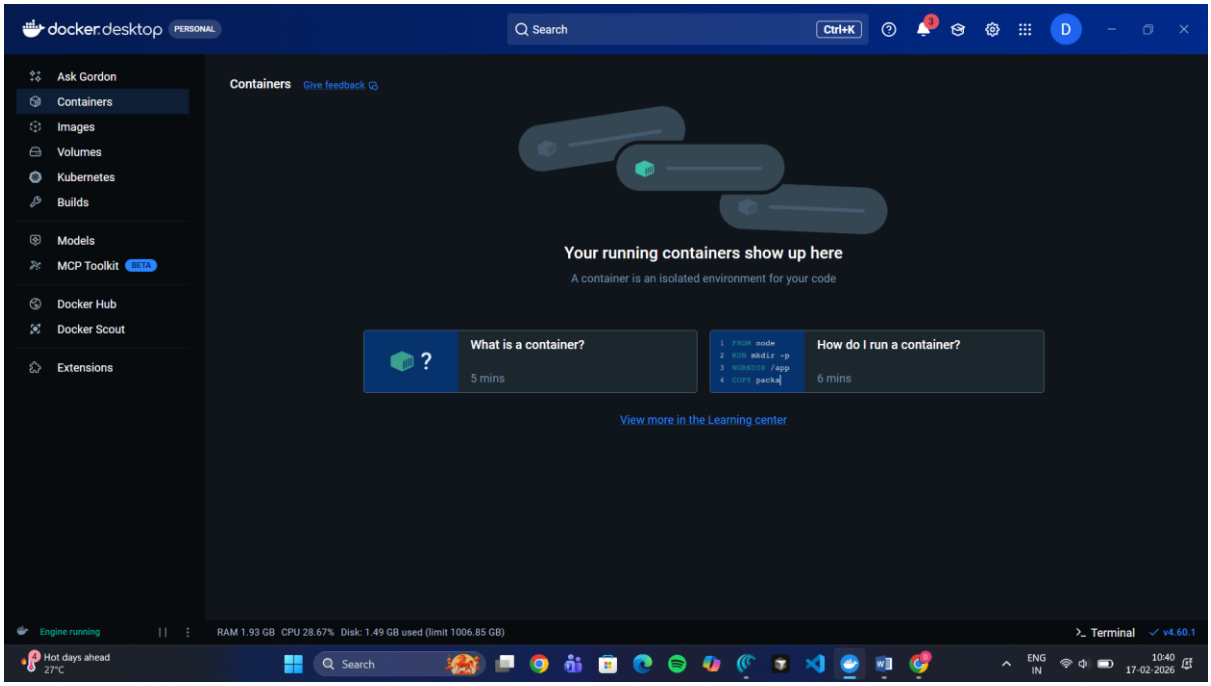

DivyanshSingh8899/College-M
DevOps-Automation Config - J
DevOps-Automation #2 - Jenki
Jenkins Build Failure Debuggin
localhost:8080/job/DevOps-Automation/2/
Jenkins / DevOps-Automation / #2
Status
Changes
Console Output
Edit Build Information
Delete build '#2'
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
#2 (17 Feb 2026, 10:21:59)
Add description
Keep this build forever
Started by user Divyansh Singh
Started 39 sec ago
Took 9.4 sec
This run spent:
14 ms waiting;
9.4 sec build duration;
9.4 sec total from scheduled to completion.
Revision: 26f53253ddf43ff12f1e226c25d7f12b0076405c
Repository: https://github.com/DivyanshSingh8899/College-Management-System-Python.git
refs/remotes/origin/main
No changes.
REST API
Jenkins 2.541.1
Air: Moderate
Now
Search
ENG
IN
10:22
17-02-2026


DivyanshSingh8899/College-M
DevOps-Automation Config - J
Console - #2 - Jenkins
Jenkins Build Failure Debuggin
localhost:8080/job/DevOps-Automation/2/console
Jenkins / DevOps-Automation / #2
Status
Changes
Console Output
Edit Build Information
Delete build '#2'
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as plain text
Started by user Divyansh Singh
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/DivyanshSingh8899/College-Management-System-Python.git
> git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation # timeout=10
Fetching upstream changes from https://github.com/DivyanshSingh8899/College-Management-System-Python.git
> git.exe --version # timeout=10
> git --version # 'git version 2.47.0.windows.2'
> git.exe fetch --tags --force --progress -- https://github.com/DivyanshSingh8899/College-Management-System-Python.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe config remote.origin.url https://github.com/DivyanshSingh8899/College-Management-System-Python.git # timeout=10
> git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 26f53253ddf43ff12f1e226c25d7f12b0076405c (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 26f53253ddf43ff12f1e226c25d7f12b0076405c # timeout=10
> git.exe branch -a -v --no-abbrev # timeout=10
> git.exe checkout -b main 26f53253ddf43ff12f1e226c25d7f12b0076405c # timeout=10
Commit message: "Resolve merge conflict in faculty.py"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Check Python)
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation>where python
C:\Python313\python.exe
Air: Moderate
Now
Search
ENG
IN
10:23
17-02-2026


DivyanshSingh8899/College-M
DevOps-Automation Config - J
Console - #2 - Jenkins
Jenkins Build Failure Debuggin
localhost:8080/job/DevOps-Automation/2/console
Jenkins
DevOps-Automation
#2
> git.exe checkout -b main 26f53253ddf43ff12f1e226c25d7f12b0076405c # timeout=10
Commit message: "Resolve merge conflict in faculty.py"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Check Python)
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation>where python
C:\Python313\python.exe
C:\Program Files\Python312\python.exe
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation>python --version
Python 3.13.2
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Install Dependencies)
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation>python -m pip install --upgrade pip
Requirement already satisfied: pip in C:\Python313\Lib\site-packages (26.0.1)
C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation>if exist requirements.txt (python -m pip install -r requirements.txt) else (echo No requirements.txt found)
No requirements.txt found
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.541.1
Air: Moderate
Now
Search
ENG
IN
10:23
17-02-2026


File Edit Selection View Go Run Terminal Help
College-Management-System-Python
EXPLORER
COLLEGE-MANAGEMENT-...
__pycache__
data
frontend
.gitignore
Alumni.py
Course.py
Dockerfile
faculty.py
Fees.py
main.py
Student.py
Course.py
Dockerfile
FROM python:3.10-slim
WORKDIR /app
COPY . .
RUN if [-f requirements.txt]; then pip install --no-cache-dir -r requirements.txt; fi
CMD ["python", "main.py"]
PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
powershell
PS C:\Users\hp\OneDrive\Desktop\Agile&Devops\College-Management-System-Python>
OUTLINE
TIMELINE
CHAT
RECENT SESSIONS
Pushing changes to College Mana...
Completed
Local • 23 hrs
Show More
Build with Agent
AI responses may be inaccurate.
Generate Agent Instructions to onboard AI onto your codebase.
Dockerfile
Describe what to build next
Agent
Auto
main
CodeTogether
Ln 10, Col 1
Spaces: 4
UTF-8
CRLF
Docker
Go Live
Trending videos
Minions & Mons...
Search
ENG
IN
10:24
17-02-2026






```
stage('Build Docker Image') {
    steps {
        bat 'docker build -t college-system:latest .'
```

Started by user Divyansh Singh

```
[Pipeline] Start of Pipeline

[Pipeline] node

Running on Jenkins

 in C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Checkout)

[Pipeline] git

The recommended git tool is: NONE

No credentials specified

 > git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation\.git # timeout=10

Fetching changes from the remote Git repository

 > git.exe config remote.origin.url https://github.com/DivyanshSingh8899/College-Management-System-Python.git # timeout=10

Fetching upstream changes from https://github.com/DivyanshSingh8899/College-Management-System-Python.git

 > git.exe --version # timeout=10

 > git --version # 'git version 2.47.0.windows.2'

 > git.exe fetch --tags --force --progress -- https://github.com/DivyanshSingh8899/College-Management-System-Python.git +refs/heads/*:refs/remotes/origin/* # timeout=10

 > git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10

Checking out Revision 26f53253ddf43ff12f1e226c25d7f12b0076405c (refs/remotes/origin/main)

 > git.exe config core.sparsecheckout # timeout=10

 > git.exe checkout -f 26f53253ddf43ff12f1e226c25d7f12b0076405c # timeout=10

 > git.exe branch -a -v --no-abbrev # timeout=10

 > git.exe branch -D main # timeout=10

 > git.exe checkout -b main 26f53253ddf43ff12f1e226c25d7f12b0076405c # timeout=10

Commit message: "Resolve merge conflict in faculty.py"

 > git.exe rev-list --no-walk 26f53253ddf43ff12f1e226c25d7f12b0076405c # timeout=10

[Pipeline] }

[Pipeline] // stage
```

```
[Pipeline] stage
[Pipeline] { (Check Python)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation>where python
C:\Python313\python.exe
C:\Program Files\Python312\python.exe
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation>python --version
Python 3.13.2
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Install Dependencies)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation>python -m pip install --upgrade pip
Requirement already satisfied: pip in C:\Python313\Lib\site-packages (26.0.1)

C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation>if exist requirements.txt (python -m pip install -r requirements.txt )  else (echo No requirements.txt found )
No requirements.txt found
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build Docker Image)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\DevOps-Automation>docker build -t college-system:latest .
```

#0 building with "default" instance using docker driver

#1 [internal] load build definition from Dockerfile

#1 transferring dockerfile:

#1 transferring dockerfile: 210B 0.1s done

#1 DONE 0.2s

#2 [internal] load metadata for docker.io/library/python:3.10-slim

#2 DONE 4.4s

#3 [internal] load .dockerignore

#3 transferring context: 2B done

#3 DONE 0.1s

#4 [internal] load build context

#4 DONE 0.0s

#5 [1/4] FROM docker.io/library/python:3.10-slim@sha256:e508a34e5491225a76fbb9e0f43ebde1f691c6a689d096d7510cf7fb17d4ba6f

#5 resolve docker.io/library/python:3.10-slim@sha256:e508a34e5491225a76fbb9e0f43ebde1f691c6a689d096d7510cf7fb17d4ba6f 0.1s done

#5 DONE 0.5s

#4 [internal] load build context

#4 transferring context: 174.05kB 1.0s done

#4 DONE 1.1s

#5 [1/4] FROM docker.io/library/python:3.10-slim@sha256:e508a34e5491225a76fbb9e0f43ebde1f691c6a689d096d7510cf7fb17d4ba6f

#5 sha256:63ebb7511fc188c7eca8836db8cac7bc37e7d188bd4e5fc617fa0f20979d8a7e 0B / 250B 0.8s

#5 sha256:690eaffcf0e9a6e579bf82062d0d78590bd1bc000a309b8e76ff4ca460bcdb6f 0B / 1.29MB 0.6s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 0B / 13.82MB 0.6s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 0B / 29.78MB 0.6s

#5 sha256:63ebb7511fc188c7eca8836db8cac7bc37e7d188bd4e5fc617fa0f20979d8a7e 250B / 250B 0.8s done

#5 sha256:690eaffcf0e9a6e579bf82062d0d78590bd1bc000a309b8e76ff4ca460bcdb6f 1.05MB / 1.29MB 3.2s

#5 sha256:690eaffcf0e9a6e579bf82062d0d78590bd1bc000a309b8e76ff4ca460bcdb6f 1.29MB / 1.29MB 3.6s done

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 1.05MB / 13.82MB 4.1s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 2.10MB / 13.82MB 5.6s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 1.05MB / 29.78MB 5.7s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 3.15MB / 13.82MB 7.4s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 3.15MB / 29.78MB 7.7s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 4.19MB / 13.82MB 9.2s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 5.24MB / 13.82MB 10.7s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 5.24MB / 29.78MB 11.1s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 6.29MB / 13.82MB 12.3s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 7.34MB / 13.82MB 14.0s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 7.34MB / 29.78MB 14.6s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 8.39MB / 13.82MB 15.8s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 9.44MB / 13.82MB 17.3s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 9.44MB / 29.78MB 18.3s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 10.49MB / 13.82MB 18.9s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 11.53MB / 13.82MB 20.4s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 12.58MB / 13.82MB 22.1s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 11.53MB / 29.78MB 21.9s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 13.63MB / 13.82MB 23.7s

#5 sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 13.82MB / 13.82MB 24.0s done

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 13.63MB / 29.78MB 24.8s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 15.73MB / 29.78MB 26.5s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 17.83MB / 29.78MB 28.3s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 19.92MB / 29.78MB 30.0s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 22.02MB / 29.78MB 31.6s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 24.12MB / 29.78MB 33.3s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 26.21MB / 29.78MB 35.1s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 28.31MB / 29.78MB 36.9s

#5 sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 29.78MB / 29.78MB 38.2s done

#5 extracting sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0

#5 extracting sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 1.7s done

#5 DONE 40.6s

#5 [1/4] FROM docker.io/library/python:3.10-slim@sha256:e508a34e5491225a76fbb9e0f43ebde1f691c6a689d096d7510cf7fb17d4ba6f

#5 extracting sha256:690eaffcf0e9a6e579bf82062d0d78590bd1bc000a309b8e76ff4ca460bcdb6f

#5 extracting sha256:690eaffcf0e9a6e579bf82062d0d78590bd1bc000a309b8e76ff4ca460bcdb6f 0.2s done

#5 DONE 40.8s

#5 [1/4] FROM docker.io/library/python:3.10-slim@sha256:e508a34e5491225a76fbb9e0f43ebde1f691c6a689d096d7510cf7fb17d4ba6f

#5 extracting sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6

#5 extracting sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 1.1s done

#5 extracting sha256:63ebb7511fc188c7eca8836db8cac7bc37e7d188bd4e5fc617fa0f20979d8a7e

#5 extracting sha256:63ebb7511fc188c7eca8836db8cac7bc37e7d188bd4e5fc617fa0f20979d8a7e 0.1s done

#5 DONE 42.0s

#6 [2/4] WORKDIR /app

#6 DONE 0.4s

#7 [3/4] COPY . .

#7 DONE 0.2s

#8 [4/4] RUN if [-f requirements.txt]; then pip install --no-cache-dir -r requirements.txt; fi

#8 DONE 1.5s

#9 exporting to image

#9 exporting layers

#9 exporting layers 0.5s done

#9 exporting manifest sha256:0e135c95d901f4a282b52e7bd78c9e701512e0e39ec27afa551720a4c198be70 0.0s done

#9 exporting config sha256:60e4eb5916bc704c7f3995c5a24fe0a99298f48ed5158e401995e2694ef6b286 0.0s done

#9 exporting attestation manifest sha256:cc4aa8f78a40eef18168ddfe071016196139fb96118fd680149caec6f8e524fc

#9 exporting attestation manifest
sha256:cc4aa8f78a40eef18168ddfe071016196139fb96118fd680149caec6f8e524fc 0.1s done

#9 exporting manifest list
sha256:a2aeec7666d2a23721da5168e48ff8440ecd78c584de5be3420a9a8d842ebbd4 0.1s done

#9 naming to docker.io/library/college-system:latest

#9 naming to docker.io/library/college-system:latest 0.0s done

#9 unpacking to docker.io/library/college-system:latest

#9 unpacking to docker.io/library/college-system:latest 0.4s done

#9 DONE 1.4s

[Pipeline] }

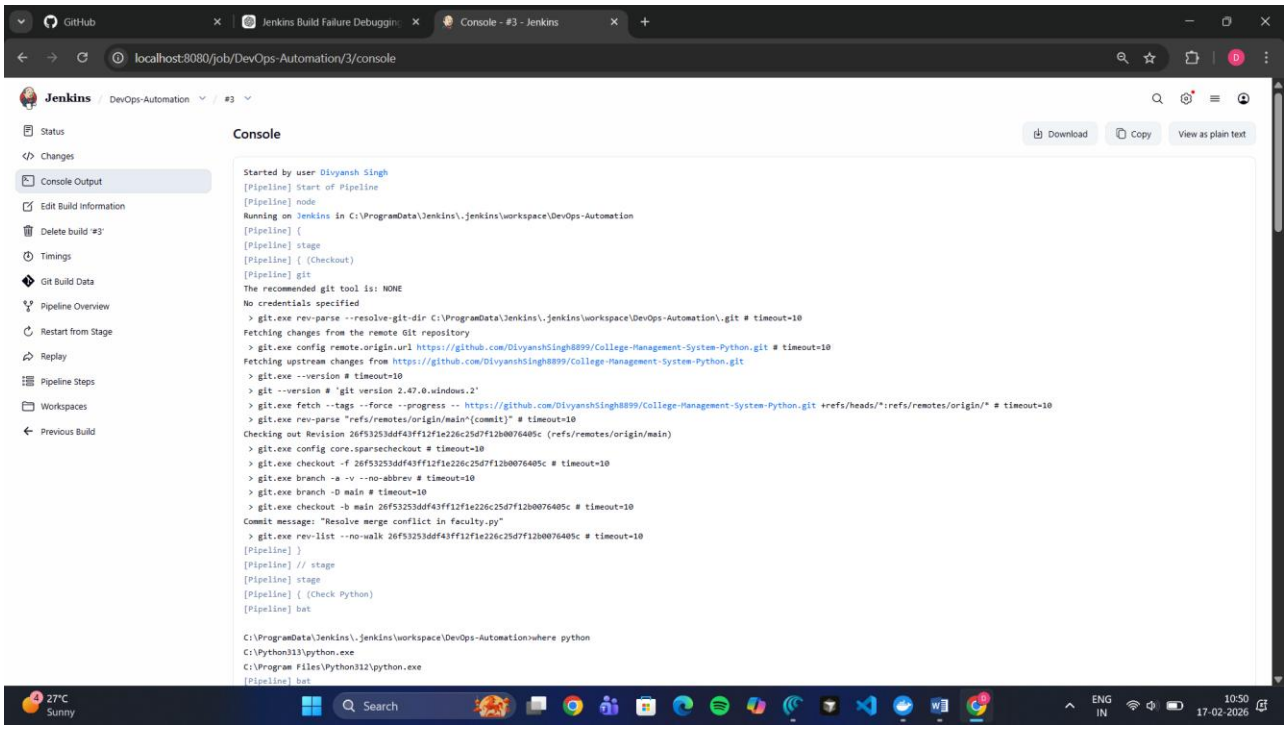
[Pipeline] // stage

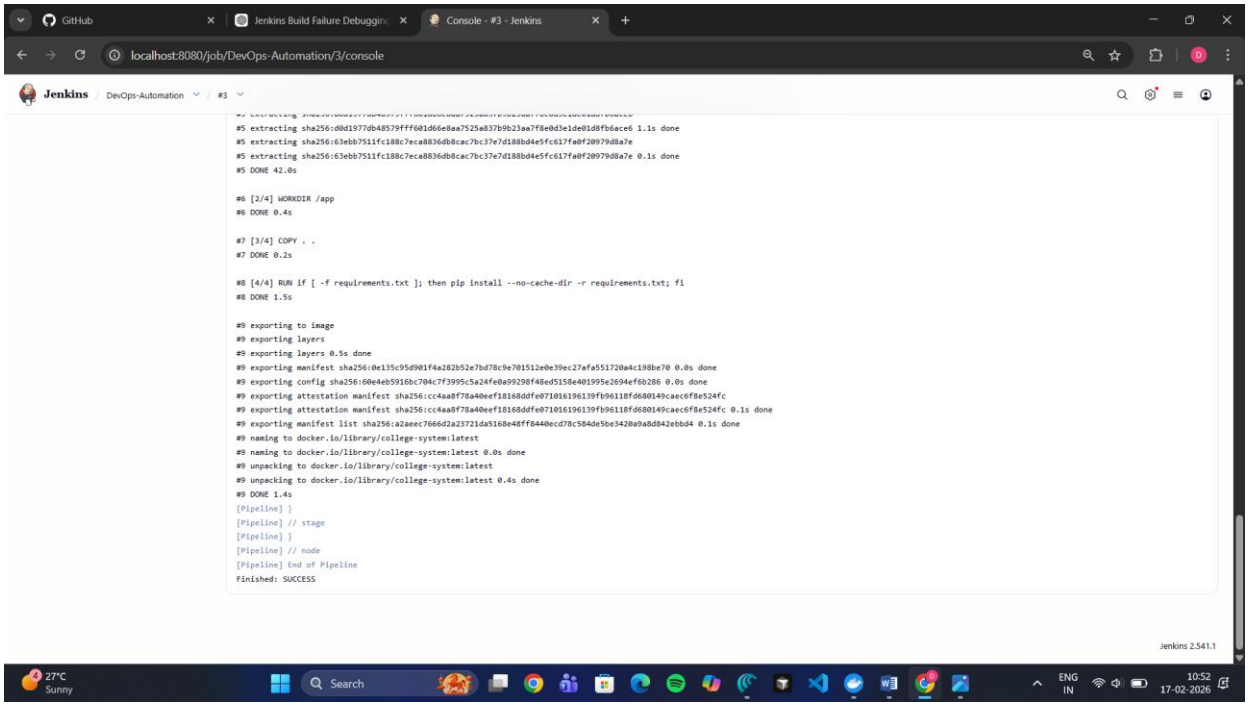
[Pipeline] }

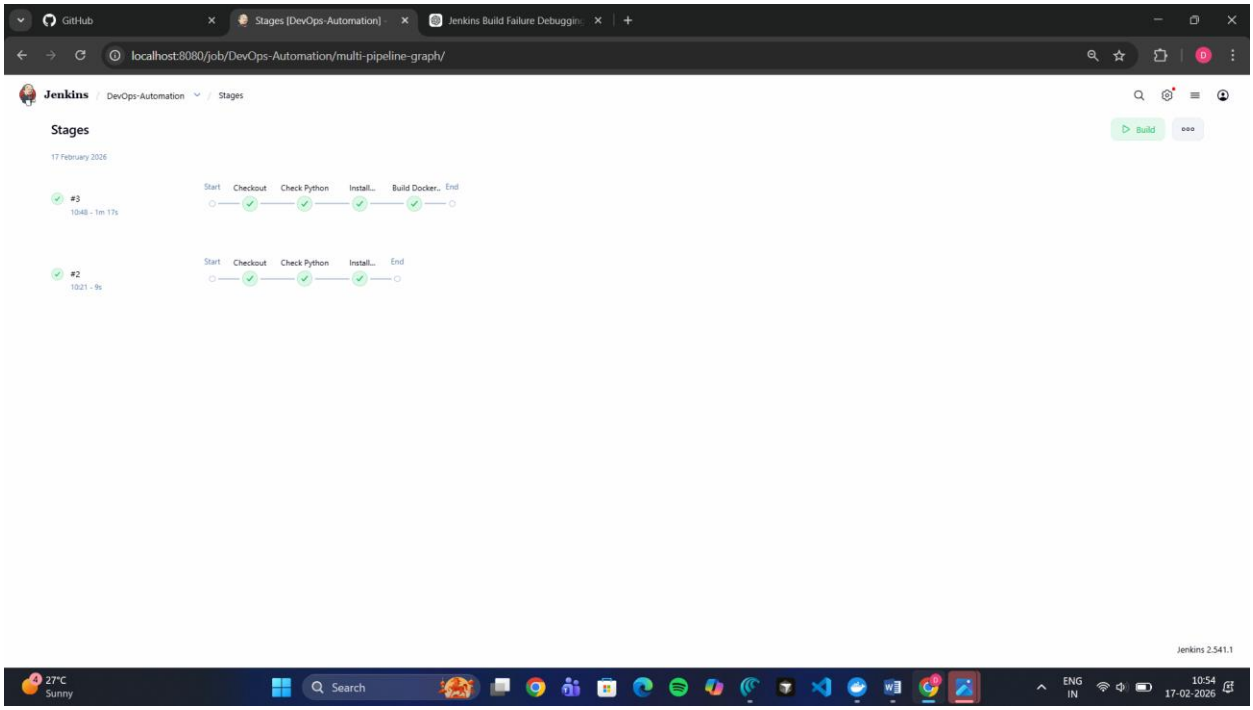
[Pipeline] // node

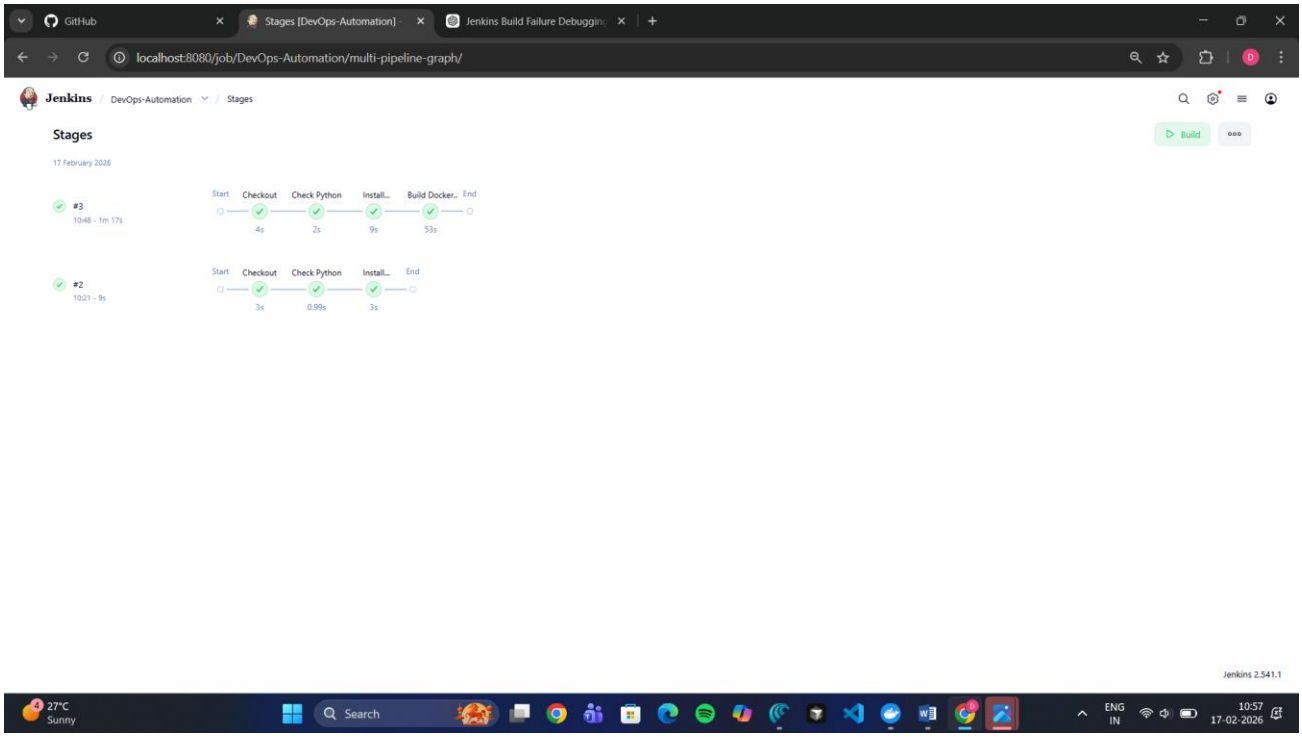
[Pipeline] End of Pipeline

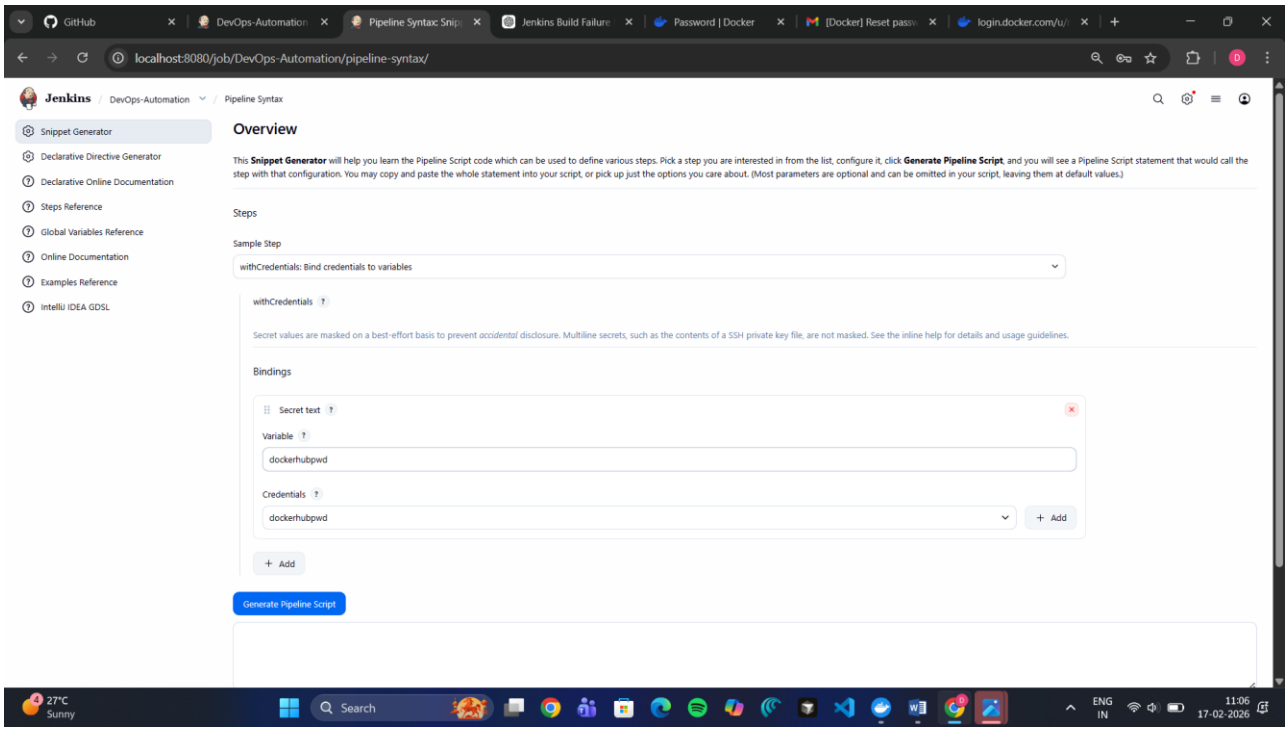
Finished: SUCCESS

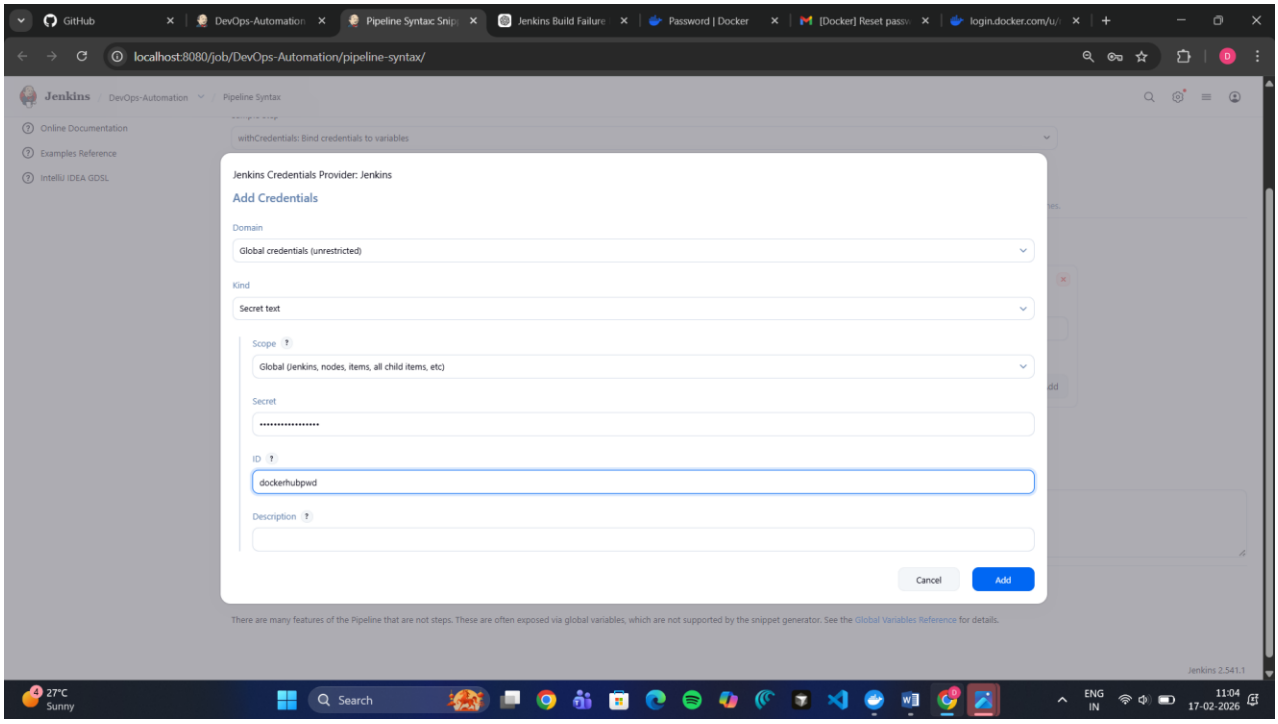



GitHub
Jenkins Build Failure Debugging
Console - #3 - Jenkins
localhost:8080/job/DevOps-Automation/3/console
Jenkins
DevOps-Automation
#3
#5 extracting sha256:d0d1977db48579fff601d66e8aa7525a837b9b23aa7f8e0d3e1de01d8fb6ace6 1.1s done
#5 extracting sha256:63ebb7511fc188c7eca8836db8cac7bc37e7d188bd4e5fc617fa0f20979d8a7e
#5 extracting sha256:63ebb7511fc188c7eca8836db8cac7bc37e7d188bd4e5fc617fa0f20979d8a7e 0.1s done
#5 DONE 42.0s
#6 [2/4] WORKDIR /app
#6 DONE 0.4s
#7 [3/4] COPY . .
#7 DONE 0.2s
#8 [4/4] RUN if [-f requirements.txt]; then pip install --no-cache-dir -r requirements.txt; fi
#8 DONE 1.5s
#9 exporting to image
#9 exporting layers
#9 exporting layers 0.5s done
#9 exporting manifest sha256:0e135c95d901f4a282b52e7bd78c9e701512e0e39ec27afa551720a4c198be70 0.0s done
#9 exporting config sha256:60e4eb5916bc704c7f3995c5a24fe0a99298f48ed5158e401995e2694ef6b286 0.0s done
#9 exporting attestation manifest sha256:cc4aa8f78a40eef18168ddfe071016196139fb96118fd680149caec6f8e524fc
#9 exporting attestation manifest sha256:cc4aa8f78a40eef18168ddfe071016196139fb96118fd680149caec6f8e524fc 0.1s done
#9 exporting manifest list sha256:a2aeec7666d2a23721da5168e48ff8440ecd78c584de5be3420a9a8d842ebbd4 0.1s done
#9 naming to docker.io/library/college-system:latest
#9 naming to docker.io/library/college-system:latest 0.0s done
#9 unpacking to docker.io/library/college-system:latest
#9 unpacking to docker.io/library/college-system:latest 0.4s done
#9 DONE 1.4s
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.541.1
27°C
Sunny
Search
ENG
IN
10:52
17-02-2026


GitHub
Stages [DevOps-Automation]
Jenkins Build Failure Debugging
localhost:8080/job/DevOps-Automation/multi-pipeline-graph/
Jenkins
DevOps-Automation
Stages
Stages
17 February 2026
Build
#3
10:48 - 1m 17s
Start
Checkout
Check Python
Install...
Build Docker...
End
#2
10:21 - 9s
Start
Checkout
Check Python
Install...
End
Jenkins 2.541.1
27°C
Sunny
Search
ENG
IN
10:54
17-02-2026


GitHub
Stages [DevOps-Automation]
Jenkins Build Failure Debuggin
localhost:8080/job/DevOps-Automation/multi-pipeline-graph/
Jenkins
DevOps-Automation
Stages
Stages
17 February 2026
Build
#3
10:48 - 1m 17s
Start
Checkout
Check Python
Install...
Build Docker...
End
4s
2s
9s
53s
#2
10:21 - 9s
Start
Checkout
Check Python
Install...
End
3s
0.99s
3s
Jenkins 2.541.1
27°C
Sunny
Search
ENG
IN
10:57
17-02-2026


GitHub
DevOps-Automation
Pipeline Syntax: Snip
Jenkins Build Failure
Password | Docker
[Docker] Reset passw
login.docker.com/u/
localhost:8080/job/DevOps-Automation/pipeline-syntax/
Jenkins
DevOps-Automation
Pipeline Syntax
Snippet Generator
Declarative Directive Generator
Declarative Online Documentation
Steps Reference
Global Variables Reference
Online Documentation
Examples Reference
IntelliJ IDEA GDSL
Overview
This Snippet Generator will help you learn the Pipeline Script code which can be used to define various steps. Pick a step you are interested in from the list, configure it, click Generate Pipeline Script, and you will see a Pipeline Script statement that would call the step with that configuration. You may copy and paste the whole statement into your script, or pick up just the options you care about. (Most parameters are optional and can be omitted in your script, leaving them at default values.)
Steps
Sample Step
withCredentials: Bind credentials to variables
withCredentials
Secret values are masked on a best-effort basis to prevent accidental disclosure. Multiline secrets, such as the contents of a SSH private key file, are not masked. See the inline help for details and usage guidelines.
Bindings
Secret text
Variable
dockerhubpwd
Credentials
dockerhubpwd
Add
Add
Generate Pipeline Script
27°C
Sunny
Search
ENG
IN
11:06
17-02-2026



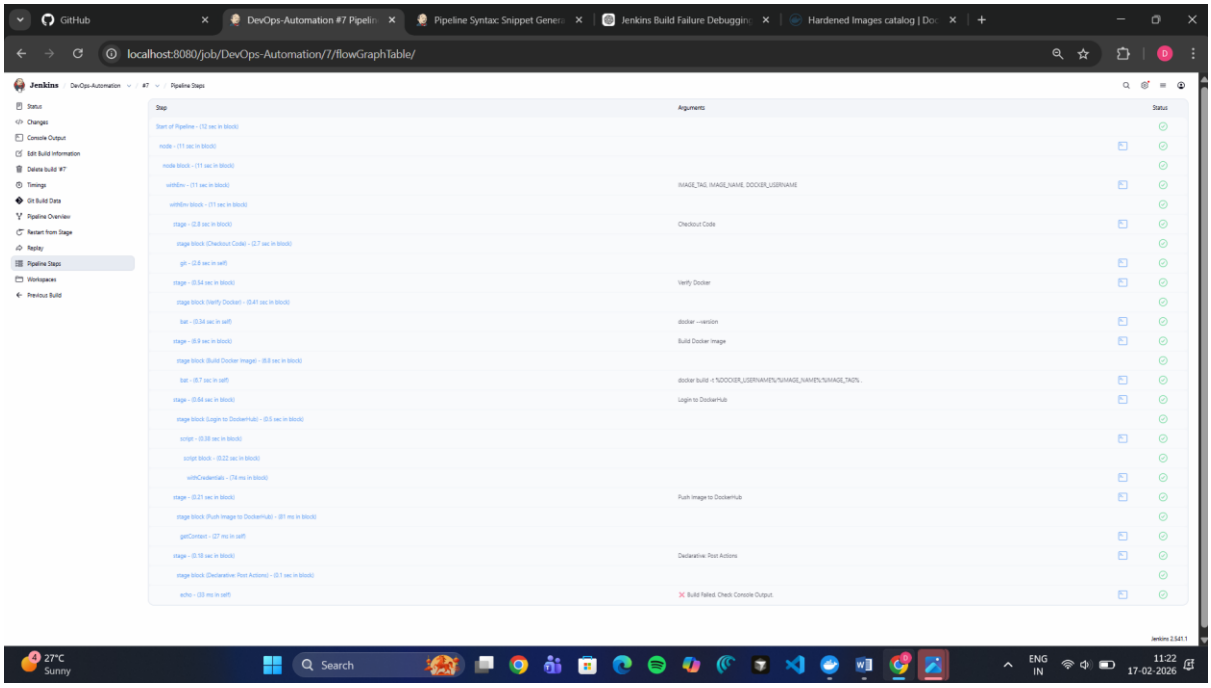
Script

```
            bat 'docker tag college-system:latest Divyansh_Singh/college-system:latest'
        }
    }

    stage('Push Image to DockerHub') {
        steps {
            script {
                withCredentials([string(credentialsId: 'dockerhubpwd', variable: 'DOCKER_PASSWORD')]) {
                    bat '''
                    echo %DOCKER_PASSWORD% | docker login -u Divyansh_Singh --password-stdin
                    docker push Divyansh_Singh/college-system:latest
                    '''
                }
            }
        }
```

Use Groovy Sandbox

Pipeline Syntax


GitHub
DevOps-Automation #7 Pipelin
Pipeline Syntax: Snippet Genera
Jenkins Build Failure Debuggin
Hardened Images catalog | Doc
localhost:8080/job/DevOps-Automation/7/flowGraphTable/
Jenkins
DevOps-Automation
#7
Pipeline Steps
Status
Changes
Console Output
Edit Build Information
Delete build '#7'
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
Step
Arguments
Status
Start of Pipeline - (12 sec in block)
node - (11 sec in block)
node block - (11 sec in block)
withEnv - (11 sec in block)
IMAGE_TAG; IMAGE_NAME; DOCKER_USERNAME
withEnv block - (11 sec in block)
stage - (2.8 sec in block)
Checkout Code
stage block (Checkout Code) - (2.7 sec in block)
git - (2.6 sec in self)
stage - (0.54 sec in block)
Verify Docker
stage block (Verify Docker) - (0.41 sec in block)
bat - (0.34 sec in self)
docker --version
stage - (6.9 sec in block)
Build Docker Image
stage block (Build Docker Image) - (6.8 sec in block)
bat - (6.7 sec in self)
docker build -t %DOCKER_USERNAME%/%IMAGE_NAME%:%IMAGE_TAG% .
stage - (0.64 sec in block)
Login to DockerHub
stage block (Login to DockerHub) - (0.5 sec in block)
script - (0.38 sec in block)
script block - (0.22 sec in block)
withCredentials - (74 ms in block)
stage - (0.21 sec in block)
Push Image to DockerHub
stage block (Push Image to DockerHub) - (81 ms in block)
getContext - (27 ms in self)
stage - (0.10 sec in block)
Declarative: Post Actions
stage block (Declarative: Post Actions) - (0.1 sec in block)
echo - (33 ms in self)
Build Failed. Check Console Output.
Jenkins 2.541.1
27°C
Sunny
Search
ENG
IN
11:22
17-02-2026